Evolutionary Dynamics : PH 4209

The Trailer

PH4209: Topics

- ❖ Introduction to Evolution
- ❖ Concepts of sequence Spaces and Fitness Landscapes
- ❖ Evolutionary Models
- ❖ Evolutionary Games

- ❑ Effect of Finite Populations
- ❑ Evolutionary Games in Finite Populations

- ➢ Introduction to Evolutionary Graph Theory
- ➢ Spatial Games: Cooperation and Conflict between interacting agents
- ➢ Evolutionary Games on Networks

- ✓ Evolutionary Dynamics of Cancer OR Role of Learning in Evolution

Objective

To develop a quantitative understanding of evolutionary processes

Tools

Basic Calculus and basic statistical concepts, simple computer programs

Books

Evolutionary Dynamics by Martin Nowak (Primary)

Evolution and the Theory of Games by John Maynard Smith (Secondary)

Mathematical Models of Social Evolution: A Guide for the Perplexed by McElreath & Boyd

Evolutionary Games and Population Dynamics by Hofbauer & Sigmund

Grading

Marks based on assignments, mid-term exams, final exams, paper/proposal presentation

Nothing in Biology makes sense except in the light of evolution.

----- Theodosius Dobzhansky

Characteristics of a living system

A living system is subject to

- Changes brought about by mutations in their genome
- Competition with other variants in the population
- Mutations provide the source of variation by allowing new variants (mutants) to emerge.
- Selection acts on those variants and can preferentially select one type over the other

Darwinian Evolution

- Mutations occur at ***random***

  Mutations can be advantageous, deleterious or neutral.

- A *random* mutation can confer a slight *selective advantage* to an individual which enhances its ability to survive and reproduce in the given environment with respect to its competitors.

- This slight advantage leads to its progeny surviving and propagating with greater probability.

- Fixation of the mutant - Over time this can lead to the mutant individual taking over the entire population.

The Peppered Moth Story (Pictures taken by Olaf Leillinger)

Pre-industrial revolution

Post-industrial revolution

Post pollution control

Big Questions

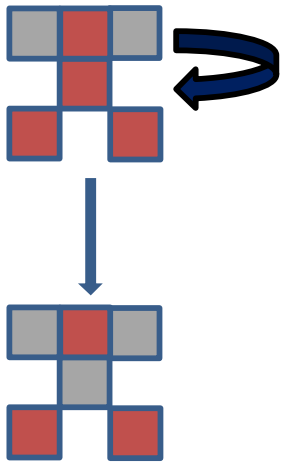
- How do populations evolve ?
- How does the ***mutation*** rates of agents in the population affect long-term population structure/composition ?
- What is the effect of ***selection*** on population structure when there are mutants with varying fitness in the population ? *(Survival of the fittest ?)*
- Under what conditions can a ***neutral*** (no selective advantage or disadvantage relative to the wild type) mutant take over the entire population by chance ?
- How did ***cooperation*** emerge in a world where individuals try to maximize their benefits and minimize their costs ?
- Role of underlying network structure on the evolutionary dynamics of the population
- What is the effect of finite population size on evolutionary dynamics ?
- How can we explain the sudden proliferation of certain infected cell types which lead to cancer ? OR What are the consequences of incorporate learning in decision-making ?

Outline

- ❖ **Evolution in *Infinite* Populations**
- ❖ **Evolution in *Finite* Populations**
- ❖ **Evolutionary Games in Finite, *Well-mixed* Populations**
- ❖ **Evolutionary Games in *Structured* Populations**

Two Examples

1. ***Random Selection:*** Evolution of population structure of two types of individuals having the same fitness.



- Two types represented by two different colours, **red** and **grey.**
- The two colours are initially randomly distributed across the grid.
- Evolution is mimicked by each patch randomly picking another patch to update its own colour
- Over time, one colour will gain a slight dominance over the other.
- The dominant colour will spread as it is more likely to be picked to update the colour of a patch.
- The random nature of the process can lead to shifts in the dominant colour
- Eventually one colour takes over the entire grid.
- Since there is no advantage of one colour over the other, each colour is equally likely to take over the entire grid.

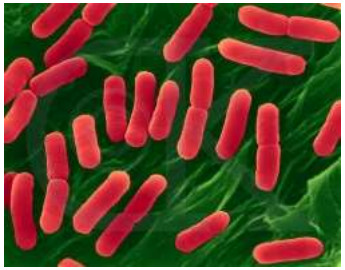
GenDrift-global.nlogo

2. ***Fitness-dependent selection :*** Evolution of a population of cooperative/altruistic and selfish agents

Altruism.nlogo

- Evolution of the population depends upon fitness determined by two phenotypic traits.
- Cooperative/altruistic behaviour (phenotype) comes with a cost as well as a benefit to the agent.
- There is no cost for selfish behaviour.
- Due to the cost associated with altruism/cooperation, the fitness of altruistic agents are somewhat **less** than the fitness of selfish agents.
- Population evolution is mimicked by changes in the colour of each agent over time.
- Colour changes are determined by the fitness and number of altruistic and selfish agents surrounding each agent.
- Population evolves from an initial state with equal numbers of altruists and selfish agents to a final stage in which either selfish or altruistic agents dominate.
- Gives insight into how cooperation/altruistic behaviour can be sustained in spite of the cost associated with such a phenotype.

Rate Equations



A: Number (population size) of ecoli cells;

a – growth rate of ecoli cells identified as fitness (a measure of reproductive success)

$$\frac{dA}{dt} = aA \quad \rightarrow \quad A(t) = A_0 \exp(at) \quad \rightarrow \text{ unbounded growth}$$

$$\frac{dA}{dt} = rA$$ r= relative growth rate = a-d ➔ unbounded growth if a>d

Unbounded growth *cannot* be sustained; nutrients required are finite, any environment has a *finite carrying capacity (K)*

$$\frac{dA}{dt} = rA(1 - A/K) \quad \rightarrow \quad A(t) = \frac{KA_0 \exp(rt)}{K + A_0(\exp(rt) - 1)}$$ As $t \rightarrow \infty$; $A(t) \rightarrow K$

How does a *heterogeneous* population evolve?

- ❖ Under what conditions does one type outcompete the other?
- ❖ Under what conditions is heterogeneity maintained in the population?

Rate Equations for Heterogenous Populations: Selection *without* mutation

A, **B**: Number (population size) of two different types;

Population Fractions: x=A/(A+B); y=B/(A+B); ***x+y=1***

a, **b** – fitness of types A and B respectively.

The sub-populations A and B are interacting; growth of one has to come at the expense of the other

$$\frac{dx}{dt} = x(a-\phi) \qquad \frac{dy}{dt} = y(b-\phi)$$

$$x+y=1 \Rightarrow \frac{dx}{dt}+\frac{dy}{dt}=0 \Rightarrow \phi = ax+by$$ ➔ ϕ : Average fitness of the population

$$\frac{dx}{dt} = x(1-x)(a-b)$$

If a>b: $\frac{dx}{dt} > 0; x(t) \to 1$: **A** outcompetes B; ***x=1*** is the stable equilibrium solution

If a<b: $\frac{dx}{dt} < 0; x(t) \to 0$: **B** outcompetes A; ***x=0*** is the stable equilibrium solution

Survival of the Fittest

Question: What happens when the fitness of A and B are equal?

Rate Equations for Heterogenous Populations: Selection *without* mutation in *Sub* and *Super*-Exponential growth models

A, **B**: Number (population size) of two different types;

Population Fractions: x=A/(A+B); y=B/(A+B); ***x+y=1***

a, **b** – fitness of types A and B respectively.

The sub-populations A and B are interacting; growth of one has to come at the expense of the other

$$\frac{dx}{dt} = axc - \phi x \qquad \frac{dy}{dt} = ayc - \phi y$$

$$x + y = 1 \Rightarrow \frac{dx}{dt} + \frac{dy}{dt} = 0 \Rightarrow \phi = axc + byc$$

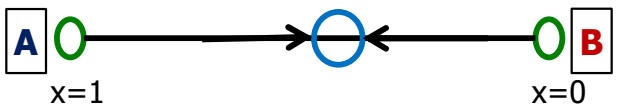
$$\frac{dx}{dt} = x(1-x)f(x) \qquad f(x) = ac^{c-1} - b(1-x)^{c-1} \qquad x_* = \frac{1}{1 + \left(\frac{a}{b}\right)^{\frac{1}{c-1}}}$$

Sub-Exponential growth: ***0 < c <1***

x= 0, 1 : *unstable* eq. states; $\mathbf{x_*}$ **:** ***stable*** eq. state

If x>$\mathbf{x_*}$: $\frac{dx}{dt} < 0; x(t) \to x_*$

If x<$\mathbf{x_*}$: $\frac{dx}{dt} > 0; x(t) \to x_*$

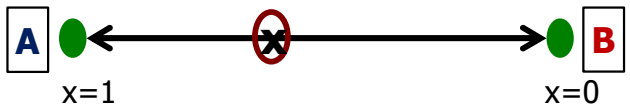


Super-Exponential growth: ***c > 1***

x= 0, 1 : ***stable*** eq. states; $\mathbf{x_*}$ **:** un*stable* eq. state

If x>$\mathbf{x_*}$: $\frac{dx}{dt} > 0; x(t) \to 1$

If x<$\mathbf{x_*}$: $\frac{dx}{dt} < 0; x(t) \to 0$



Rate Equations for Heterogenous Populations: Mutation *without* selection

u1 – mutation rate from A to B; **u2** – mutation rate from B to A

$$\frac{dx}{dt} = \text{(\# ways in which fraction of A can increase) – (\# ways in which fraction of A can decrease)}$$

$$\frac{dx}{dt} = u_2 y - u_1 x \qquad \frac{dy}{dt} = u_1 x - u_2 y$$

$$x + y = 1 \Rightarrow x_* = \frac{u_2}{u_1 + u_2}; y_* = \frac{u_1}{u_1 + u_2}$$

Equilibrium Frequencies of A and B in the population

Limiting Cases:

$$\left.\begin{aligned} u_1 >> u_2 &\Rightarrow x_* \to 0, y_* \to 1 \\ u_1 << u_2 &\Rightarrow x_* \to 1, y_* \to 0 \end{aligned}\right\}$$

Large differences in mutation rates lead to selection of one type over the other even in the *absence* of difference in fitness

General Forms of Evolution Equations

Mutation *without* selection: $\frac{dx_i}{dt} = \sum_{j=1}^{N} x_j q_{ji} - \phi x_i$

Selection *without* mutation: $\frac{dx_i}{dt} = x_i (f_i - \phi)$

$Q = [q_{ij}]$: *Mutation* matrix; q_{ij} - mutation probability from i➔j

Mutation matrix for point mutations

Hamming Distance **(h_{ij})** between 2 sequences i and j (each of length L) gives the number of positions at which the sequences differ.

u - Mutation probability at each of the L positions.

q_{ij} – Probability that replication of sequence i leads to sequence j .

$$q_{ij} = u^{h_{ij}} (1-u)^{L-h_{ij}}$$

mutations occur at **h_{ij}** positions

L-h_{ij} positions do not mutate

Consequences of mutations in viruses: Why it is difficult to find a cure for HIV?

- Viruses typically have a small genome size and RNA viruses like HIV have even smaller genomes due to higher error rates during replication

Size ~ 2000 to 2 million bases

Typical mutation rates: 10^{-3} to 10^{-5} per base/generation (RNA virus)

10^{-5} to 10^{-8} per base per generation (DNA virus)

HIV mutation rate per: u ~ $3*10^{-5}$ per base per generation

HIV genome length: L ~ 10^{4} bases

Sequence Spaces and Fitness Landscapes

$4^L \rightarrow$ Each point has 4 options ie A, G, T, C

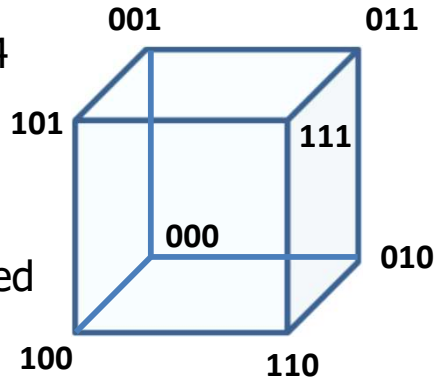
And there are L points.

Sequence Space : A sequence of length L is a point in **L-dimensional** space where each dimension has **4** discrete values corresponding to the 4 nucleotides.

There are $\mathbf{4^L}$ discrete points in the sequence space and each point corresponds to a distinct sequence of length **L**.

When only purines (A and G) and Pyrimidines (T and C) are distinguished instead of all 4 nucleotides, each of the L dimensions in sequence space has 2 discrete points ➔ the sequence space for a sequence of length L has $\mathbf{2^L}$ discrete points.

For L=3, the sequence space for sequences made up of purines (represented by 0) and pyrimidines (represented by 1) has a total of $\mathbf{2^3 = 8}$ discrete points and can be represented by the vertices of a cube.



Each sequence is connected to connected by an edge to sequences that differ from it at a single position. Example: 100 is connected to 000, 110 and 101

Movement in sequence space occurs in discrete steps and ***Evolution is a trajectory through sequence space***

Fitness Landscape: Fitness determined by the phenotype i.e. characteristics like behaviour, morphology, structure, shape etc.

Genotype ➔ Phenotype mapping determines how changes in the genome affect the phenotype – the most challenging problem in Biology

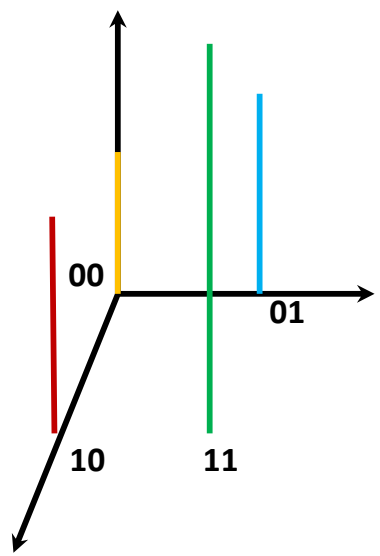
Fitness landscape : Direct mapping between ***genotype and fitness.***

An extra dim for fitness

Fitness Landscape: (L+1)dimensional space where the first L dimensions describes the sequence space and the (L+1) th dimension represents the fitness associated with each of the 4^L (or 2^L) sequences

Evolution of sequences amounts to movement in the fitness landscape and ***Adaptation is an attempt to attain peaks in the fitness landscape.***

Single Peak Fitness Landscape



Adaptation of a population of 10 sequences of length L=2

00		11		11
01		11		11
00		11		11
01	Evolution →	01	Evolution →	11
11		11		11
10		11		11
11	After some generations	10	After many more generations	11
11		11		11
10		00		11
01		11		11

Type	Frequency
00	0.2
01	0.3
10	0.2
11	0.3

Type	Frequency
00	0.1
01	0.1
10	0.1
11	0.7

Type	Frequency
00	0
01	0
10	0
11	1

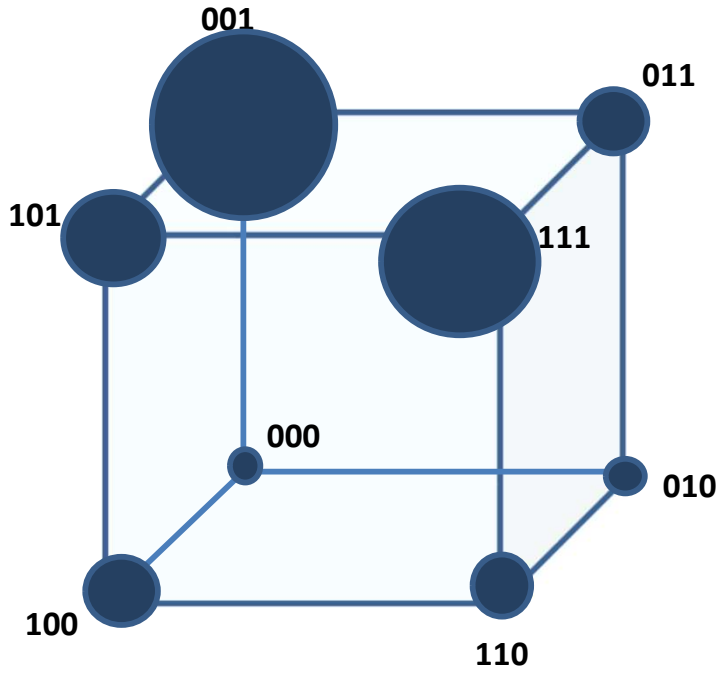
A population is well adapted if most (or all) of the members cluster around a fitness peak in the fitness landscape

Tunably rugged fitness landscape: Kauffman's NK-model (with N=3, K=1)

N=3 ➔Sequence length is 3

K=1 ➔ Fitness landscape has a *single* peak

Number of peaks in the fitness landscape



$$A \equiv [A_{ij}] = \begin{matrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{matrix}$$

Fitness of gene s_1 depends on gene s_3

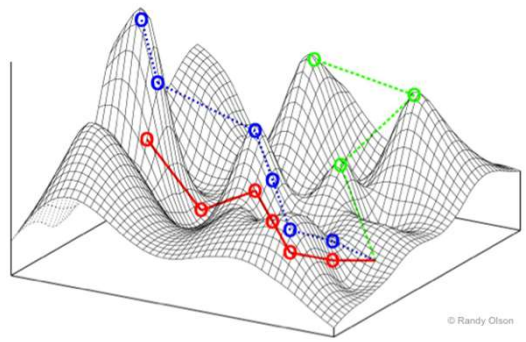
Fitness of gene s_2 depends on gene s_3

Fitness of gene s_3 depends on gene s_2

Note: A is *not* a symmetric matrix since if fitness of gene s_i depends on gene s_j it does not imply fitness of gene s_j depends on gene s_i

Example of a rugged fitness landscape with multiple peaks (K>1)

Red and Blue lines: Pathways of population evolution through mutations in gene sequences



Source: Wikipedia

Replicator/Quasi-Species Equation: Evolution with Mutation *and* selection

M. Eigen

$$\frac{dx_i}{dt} = \sum_{j=1}^{N} x_j f_j q_{ji} - \phi x_i$$

P. Schuster

Limiting Cases

Mutation *without* selection: $\frac{dx_i}{dt} = \sum_{j=1}^{N} x_j q_{ji} - \phi x_i$ when $f_i = 1$ for all i

Error-free replication: $q_{ji} = \delta_{ji} \Rightarrow \quad \frac{dx_i}{dt} = x_i (f_i - \phi)$

Useful in describing evolution of population of sequences where

- Variations between sequences arise as a result of errors during replication
- Presence of replication errors shifts the equilibrium from a pure state (survival of the fittest) to a mixed state where multiple quasi-species can coexist
- The equilibrium state does not necessarily maximize average fitness

Replicator/Quasi-Species Equation: Evolution with Mutation *and* selection

M. Eigen

P. Schuster

$$\frac{dx_i}{dt} = \sum_{j=1}^{N} x_j f_j q_{ji} - \phi x_i$$

$$\frac{d\vec{x}}{dt} = \vec{x} W - \phi \vec{x}$$

$$W = [w_{ij}] = \begin{bmatrix} f_1 & 0 & 0 & 0 \\ 0 & f_2 & 0 & 0 \\ 0 & 0 & f_3 & 0 \\ 0 & 0 & 0 & f_n \end{bmatrix} \begin{bmatrix} q_{11} & q_{12} & q_{13} & q_{1n} \\ q_{21} & q_{22} & q_{23} & q_{2n} \\ q_{31} & q_{32} & q_{33} & q_{3n} \\ q_{41} & q_{42} & q_{43} & q_{4n} \end{bmatrix} = \begin{bmatrix} f_1 q_{11} & f_1 q_{12} & f_1 q_{13} & f_1 q_{1n} \\ f_2 q_{11} & f_2 q_{22} & f_2 q_{23} & f_2 q_{2n} \\ f_3 q_{31} & f_3 q_{32} & f_3 q_{33} & f_3 q_{3n} \\ f_n q_{n1} & f_n q_{n1} & f_n q_{n3} & f_n q_{nn} \end{bmatrix}$$

Diagonal (nxn) selection matrix (nxn) mutation matrix (nxn) mutation-selection matrix

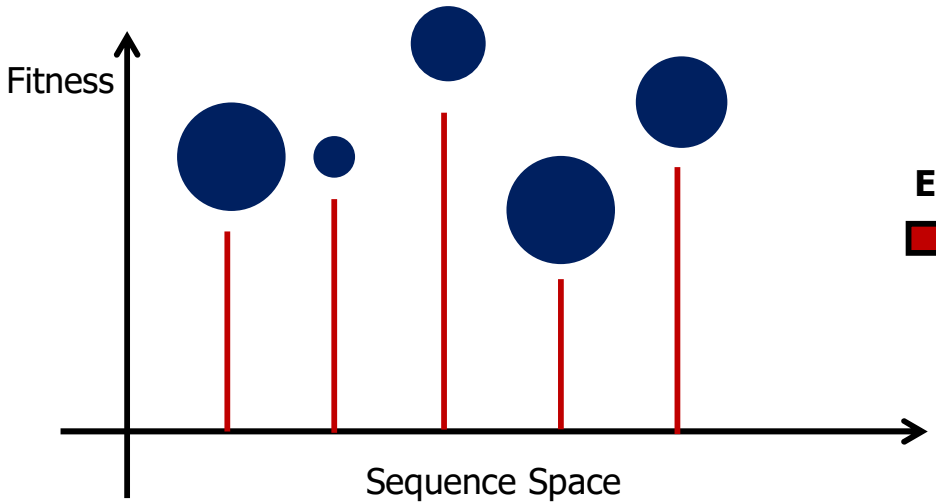
For equilibrium

Equilibrium Quasispecies Dynamics

$$\vec{x} W = \phi \vec{x}$$: ϕ is the largest eigenvalue of the mutation-selection matrix W

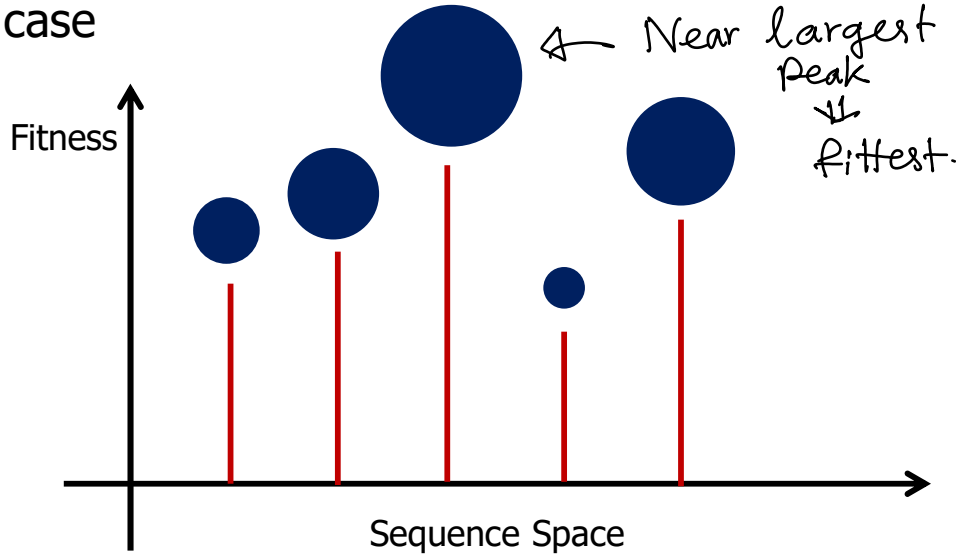
The *left-hand eigenvector* associated with this eigenvalue: *Equilibrium frequencies* of the different types making up the population, satisfying the constraint $\sum_i x_i = 1$

Adaptation: Generic case



Initial population distribution

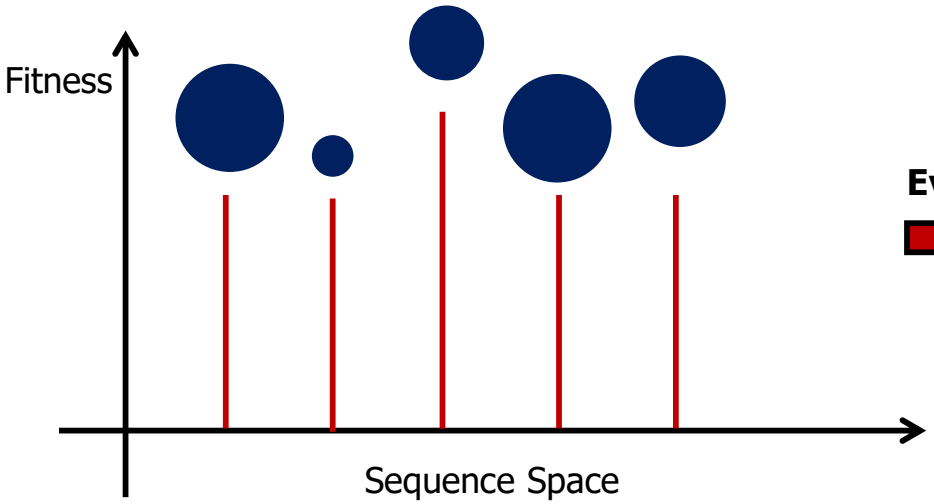
Evolution



Well adapted population clustered around the peak

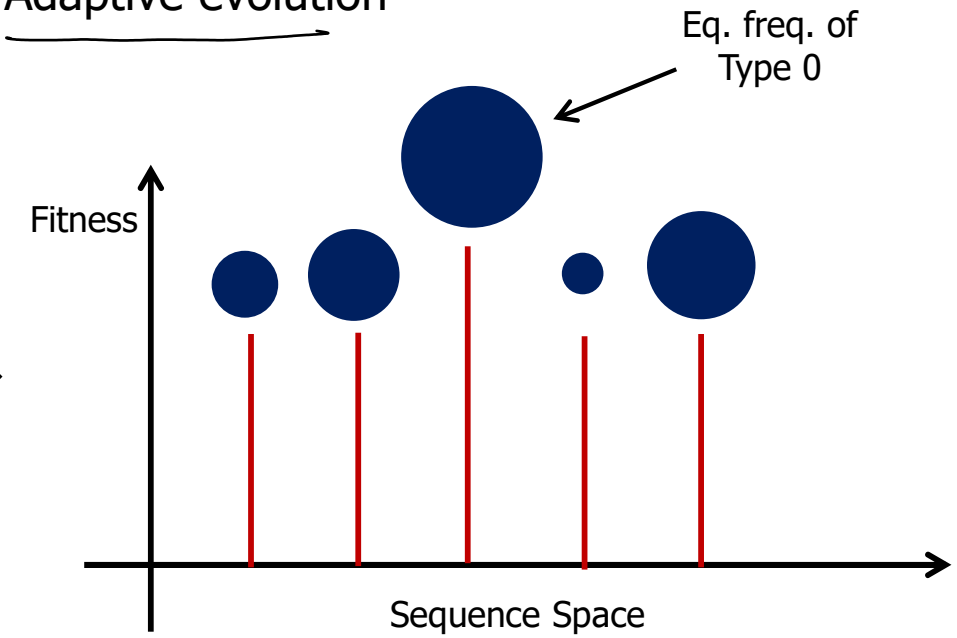
Size of the circles represent the frequency of the corresponding sequence in the population

Below Error Threshold : Adaptive evolution

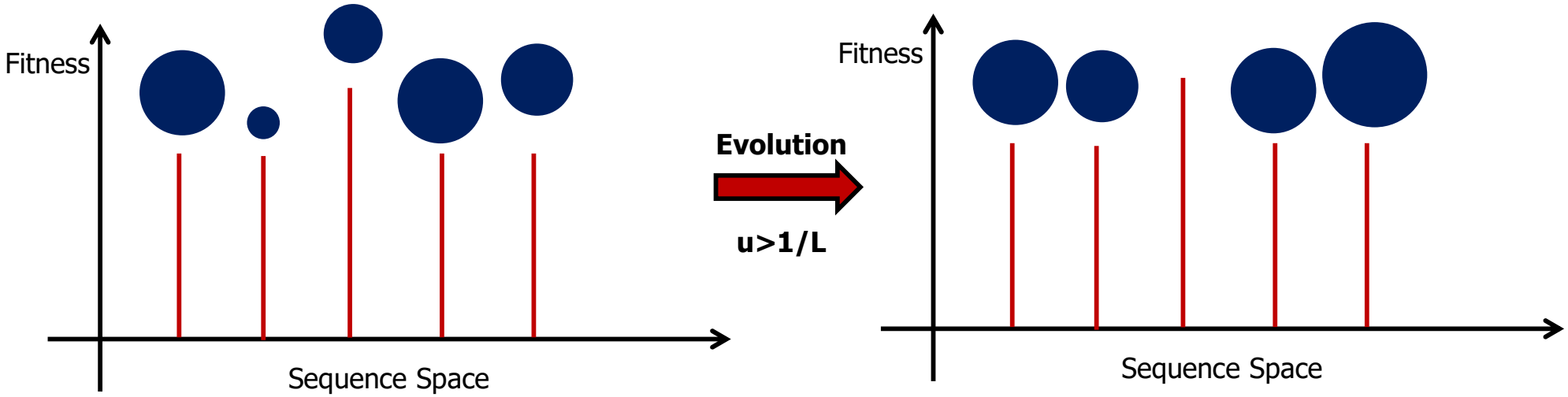


Evolution

u<1/L



Above Error Threshold : Non-adaptive evolution



Fittest Sequence Type (Type 0) lost from the population

Eigen's Paradox

- For a replicating molecule to be viable and not be subject to mutational degradation (i.e. evade the error threshold problem), *its sequence length should be small*.

  Need ← large length → Degradation } → contradiction related to stability

- For a replicating molecule to encode enzymes (i.e. to be functional), *its sequence length should be substantially large*.

How can functional molecules with long sequences survive in the population?

How does Mutation Rate per site (u) across organisms compare with the Error Threshold ($1/L$) ?

Organism	Genome length in bases	Mutation rate per base	Mutation rate per genome
RNA viruses			
Lytic viruses			
Qβ	4.2×10^3	1.5×10^{-3}	6.5
Polio	7.4×10^3	1.1×10^{-4}	0.84
VSV	1.1×10^4	3.2×10^{-4}	3.5
Flu A	1.4×10^4	7.3×10^{-6}	0.99
Retroviruses			
SNV	7.8×10^3	2.0×10^{-5}	0.16
MuLV	8.3×10^3	3.5×10^{-6}	0.029
RSV	9.3×10^3	4.6×10^{-5}	0.43
Bacteriophages			
M13	6.4×10^3	7.2×10^{-7}	0.0046
λ	4.9×10^4	7.7×10^{-8}	0.0038
T2 and T4	1.7×10^5	2.4×10^{-8}	0.0040
E. coli	4.6×10^6	5.4×10^{-10}	0.0025
Yeast (*S. cerevisiae*)	1.2×10^7	2.2×10^{-10}	0.0027
Drosophila	1.7×10^8	3.4×10^{-10}	0.058
Mouse	2.7×10^9	1.8×10^{-10}	0.49
Human (*H. sapiens*)	3.5×10^9	5.0×10^{-11}	0.16

Sources: Drake (1991, 1993) and Drake et al. (1998).

Note: Most organisms have a mutation rate per genome which is less than one, as predicted by the error threshold theory. Why Qβ and VSV have such a high mutation rate is at present unexplained.

Note: Organisms remain viable only if $uL<1$

For polio virus: $uL=0.814$

RNA virus Error Catastrophe

Anti-viral effect manifest by enhanced mutagenesis of the Polio virus genome

High mutation rates ➔ loss of viability of the Polio virus genome

Table 2. The antiviral effects of ribavirin can be directly attributed to lethal mutagenesis

	Normal	100 μM ribavirin	400 μM ribavirin	1,000 μM ribavirin
RNA-specific infectivity loss	—	3.3	18	140
Loss of total viral RNA	—	—	6	16
Total predicted titer reduction	1	3.3	100	2,200
Actual titer reduction*	1	3.2	71	2,000

*Untreated ("normal") poliovirus titer in this experiment was 1.2×10^{10} PFU per plate of HeLa cells (6×10^6 cells). Data are the average of three experiments.

Polio viruses reside near the edge of the Error Threshold

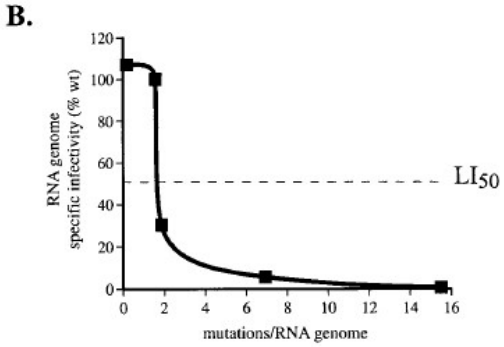
Modest (less than 2-fold) increase in mutation rate ➔ 50% of the viral population becomes unviable

4-fold increase in mutation rate ➔ 95% of the viral population becomes unviable.

Table 3. Mutation frequency in ribavirin-treated RNA virus populations

	G→A	C→T	Total mutation frequency*
Normal population	0.5	1.2	2.1
100 μM ribavirin	—	1.3	2.5
400 μM ribavirin	4.4	5.0	9.3
1,000 μM ribavirin	6.8	12.0	20.8

*Mutations per 10,000 nt sequenced.

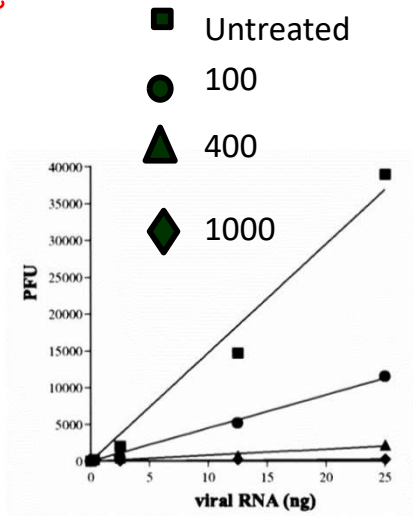


Vaccine increase the mutation rate to push it beyond threshold is then LL > 1

High mutation Rate ⇔ low infectivity.

High mutation rate produces Polio-virus mutants having low infectivity

The amount of infectious virus genomes in the population is reduced several fold as concentration of Ribavarin increases.



Ref.: Crotty, Cameron, Andino; PNAS 98, 6895-6900 (2001)

PFU: plaque forming unit

Pictorial representation of evolution by mutation without selection

we don't consider the relative fitness

A-type B-type

Time			**#A**	**#B**
Time=t	Pick Random members		**nA**	**nB**
Check type **A** or **B**	Check if RN < u1, if true mutate A to B. If False, leave A unchanged	Repeat process for other members of population		
Time=t+1			**nA-1**	**nB+1**

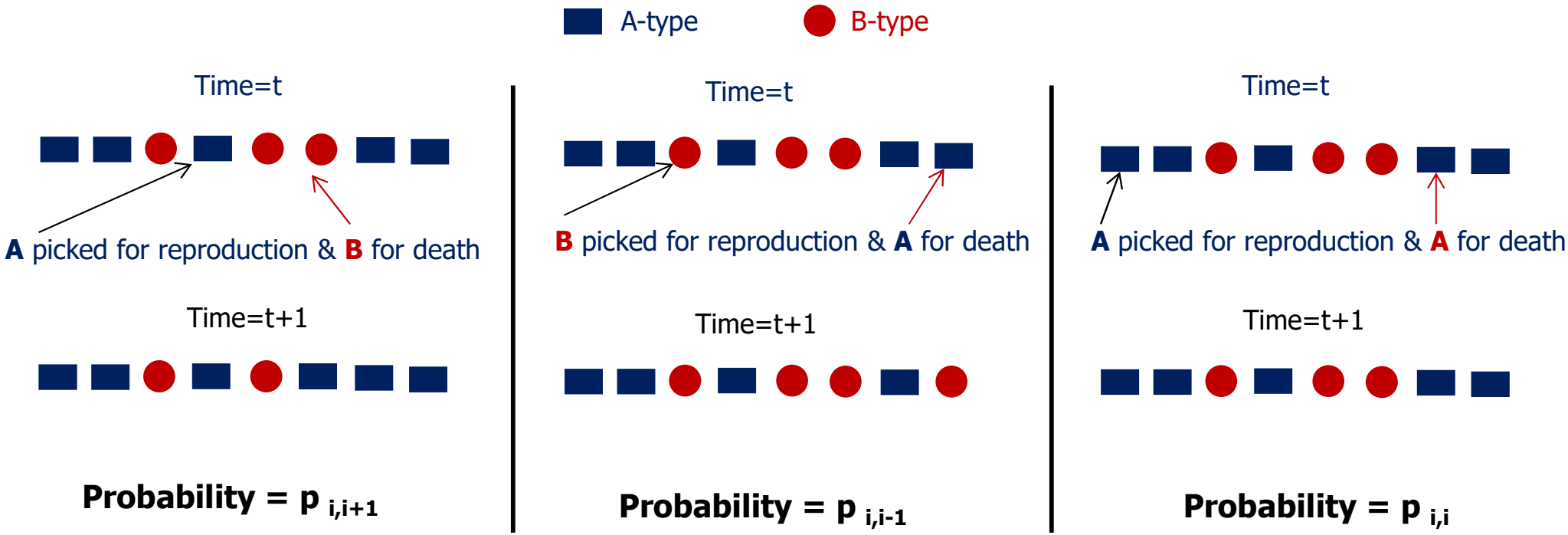
Moran Process

Assume two types of individuals A and B in the population of N individuals, both types having the *same* fitness.

In every generation, one individual is picked at random for reproduction and another individual is picked at random for death.

Question: What is the ***fixation probability*** of A ?

Fixation Probability of A: Probability that the frequency of A increases from an initial value of i/N to a final value of 1 i.e. the final population consists entirely of A



$$p_{i,i-1} + p_{i,i} + p_{i,i-1} = 1$$

Difference between Invasion and Fixation

Invasion is a special case of Fixation

In *deterministic* simulations involving the replicator equation,

Invasion by type A ➔ frequency of A increases from a very small fraction $x=\varepsilon << 1$ ➔ $x=1$

Fixation of A ➔ frequency of A increases from any x ➔ $x=1$

In *stochastic* simulations like evolution by Moran process

Invasion by type A ➔ frequency of A increases from $x=1/N$ ➔ $x=1$

Fixation of A ➔ frequency of A increases from any x (such that $0 < x=\#A/N < 1$) ➔ $x=1$

NOTE

The inability of A to *invade* a population of B does not mean that A cannot get *fixed* in the population if its initial frequency is sufficiently large.

invation is a special case of fixation

Evolution via Moran Process in Finite Populations

No. of A players = **i;** No. of B players = **N-i.** Also assume A and B has the same fitness

Prob. of choosing A for both death and reproduction = (i/N)^2

Prob. of choosing B for both death and reproduction = ((N-i)/N)^2

Prob. of choosing A for reproduction and B for death = {i/N}{(N-i)/N}= $\mathbf{p}_{\mathbf{i,i+1}}$

Prob. of choosing B for reproduction and A for death = {(N-i)/(N)}{i/N}= $\mathbf{p}_{\mathbf{i,i-1}}$

Define: $\gamma_i = \frac{p_{i,i-1}}{p_{i,i+1}} = \frac{\beta_i}{\alpha_i}$ $\quad y_i = x_i - x_{i-1}$

If x_i is the fixation probability of C's starting from a state with #C=i

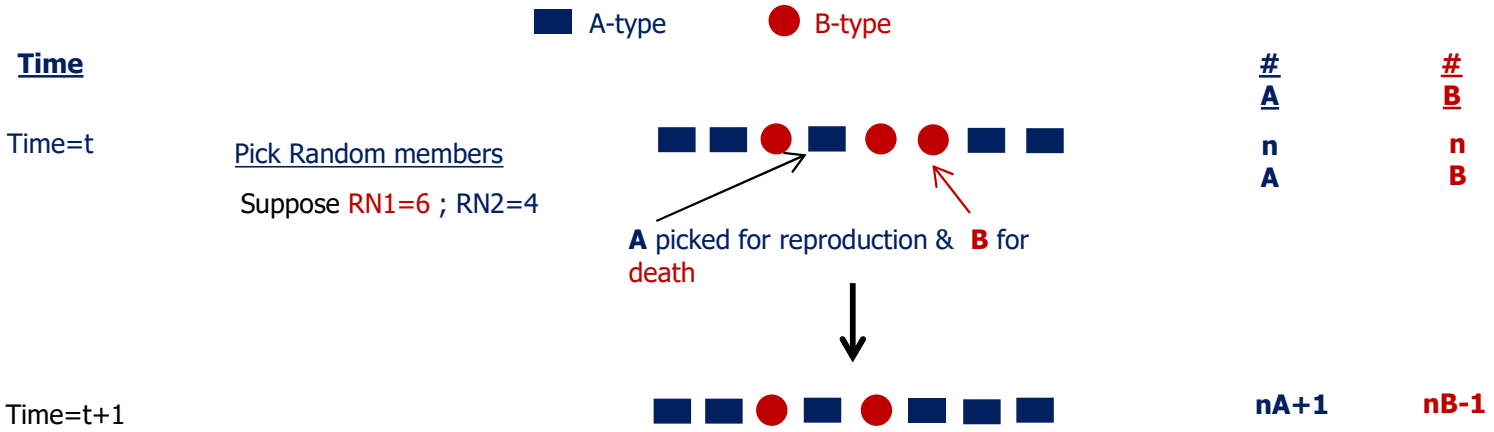
$$x_i = x_i p_{i,i} + x_{i-1} p_{i,i-1} + x_{i+1} p_{i,i+1} \Rightarrow y_{i+1} = \gamma_i y_i$$

$$\sum_{i=1}^{N} y_i = x_N - x_0 = 1$$

$$y_{i+1} = \gamma_i y_i \Rightarrow \sum_{i=1}^{N} y_i = x_1(1 + \gamma_1 + \gamma_2\gamma_1 + \ldots\ldots + \gamma_{N-1}\ldots\gamma_1) \Rightarrow x_1 \equiv \rho_C = \frac{1}{1 + \sum_{j=1}^{N-1}\prod_{k=1}^{j}\gamma_k} = \frac{1}{N}$$

$$y_{i+1} = \gamma_i y_i \Rightarrow x_i = x_1(1 + \gamma_1 + \gamma_2\gamma_1 + \ldots + \gamma_{i-1}\ldots\gamma_1) \Rightarrow x_i = \frac{1 + \sum_{j=1}^{i-1}\prod_{k=1}^{j}\gamma_k}{1 + \sum_{j=1}^{N-1}\prod_{k=1}^{j}\gamma_k} = \frac{i}{N}$$

Pictorial representation of the Simulation of Moran Process: Neutral Evolution Case



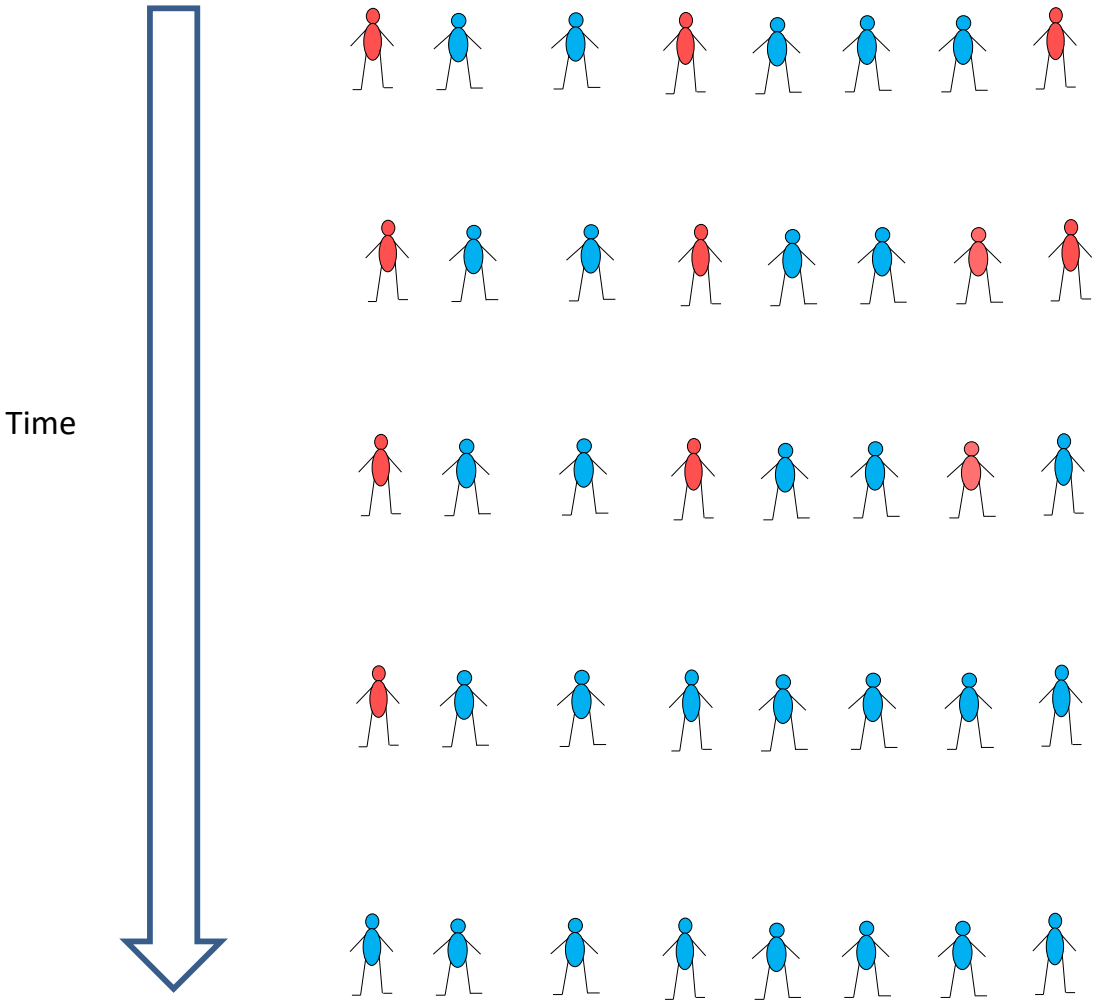
Repeat above steps as many times until #A=N or #B=N

Repeat simulation with same initial condition NT times to determine ρ_A and ρ_B

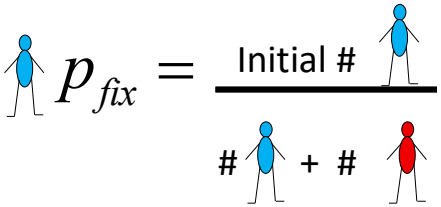
Invasion probability of A: $\rho_A \equiv x_1 = \frac{1}{N};$ Invasion probability of B: $\rho_B \equiv 1 - x_{N-1} = \frac{\prod_{k=1}^{N-1} \gamma_k}{1 + \sum_{j=1}^{N-1} \prod_{k=1}^{j} \gamma_k} = \frac{1}{N};$

For the Neutral Evolution Case, invasion probability of a single mutant is inversely proportional to population size

Neutral Evolution: A simple model without mutation



Fitness of = Fitness of



If they start with equal number of population so they have equal possiblity to be fixed in the population

Evolution with selection via Moran Process

Reproduction: Occurs with probability proportional to the fitness of the agent

Death: Occurs at random, independent of the fitness of the agent

$$p_{i,i+1} = \left(\frac{ri}{ri + 1(N-i)}\right)\left(\frac{N-i}{N}\right)$$

Prob. of picking A for reproduction

Prob. of picking B for death

$$p_{i,i-1} = \left(\frac{(N-i)1}{ri + 1(N-i)}\right)\left(\frac{i}{N}\right)$$

Prob. of picking B for reproduction

Prob. of picking A for death

Remember that, this is for Moran Process

$$\gamma_i = \frac{p_{i,i-1}}{p_{i,i+1}} = \frac{1}{r}$$

$$x_i = \frac{(1-\frac{1}{r^i})}{(1-\frac{1}{r^N})}$$

Depends on the pop. size

For r> 1; as N➔ ∞ $\quad x_1 \equiv \rho_A \simeq 1 - \frac{1}{r}$

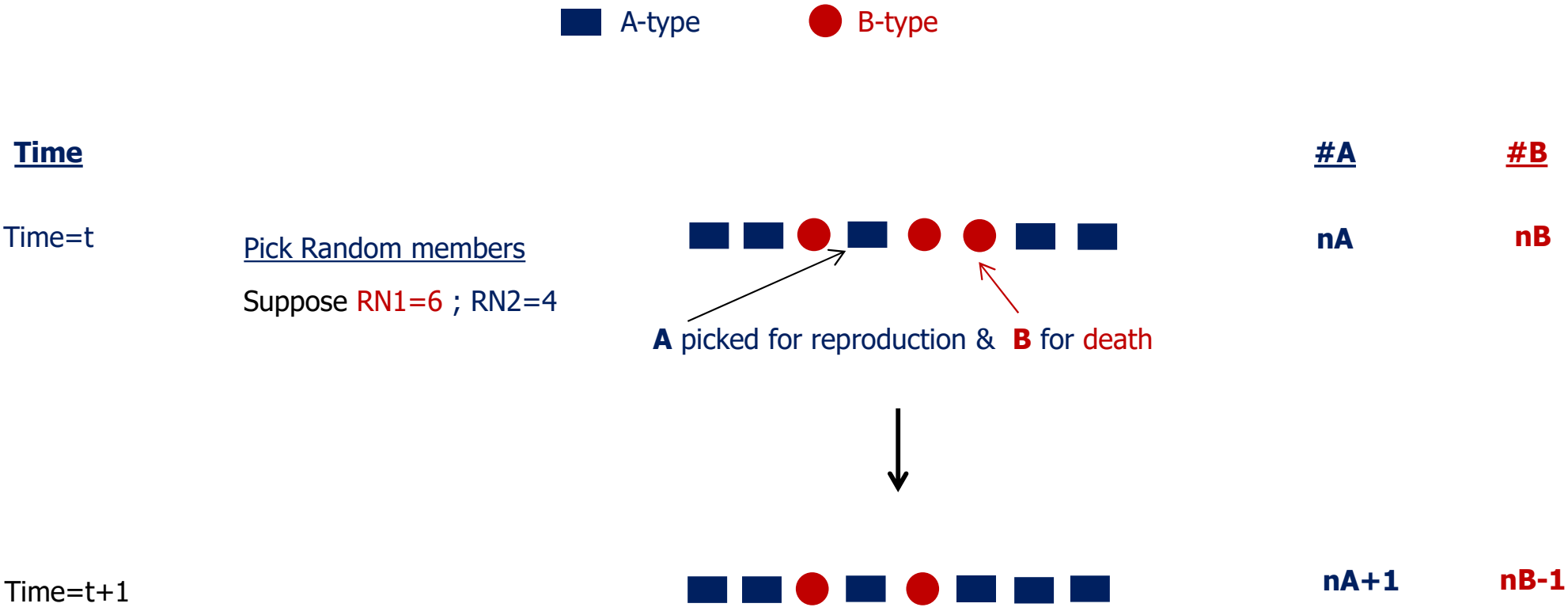
Invasion Probabilities for N=100

r	$x_1 \equiv \rho_A$
2	0.5
1.01	0.016
1	0.01

r	$x_1 \equiv \rho_A$
0.99	0.0058
0.9	0.000003

➔ Invasion by a selectively advantageous mutant is not guaranteed even for large population sizes

Pictorial representation of the Simulation of Moran Process: Neutral Evolution Case



Pictorial representation of selection without mutation: Fixation of an ad/disadvantageous mutant

Evolution by Moran Process

Time

A-type B-type

Time=1

Pick a Random member

Suppose RN=3

Parent pop. Array (P)

B picked for reproduction

A picked for death at random

Not OK

Compare with <fitness> for reproduction: Pick RN between 0&1. If RN < normalized_fitness of B, then OK

selected to be reproduced based on the prob. of fitness

OK

B becomes the seventh member of the Offspring pop. at t=2

Time=1

Offspring pop. Array (O)

B becomes the seventh member of the Offspring pop. at t=1

Offspring population at t=1 becomes the parent population t=2

Time=2

Parent pop. Array (P)

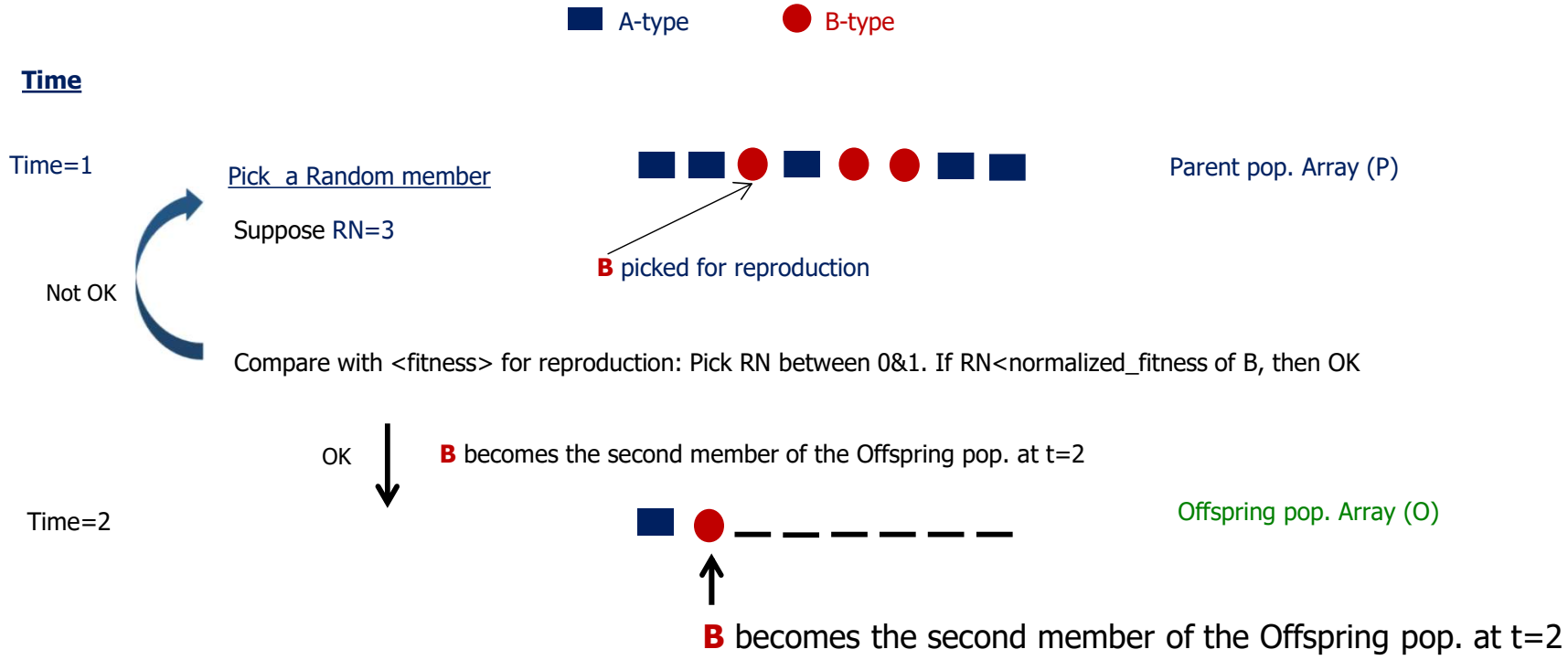
Repeat above steps to generate the parent population t=3 and so on

Note: If fitness of A is r1 and fitness of B is r2,

normalized_fitness_of_**A**=r1/(r1+r2); normalized_fitness_of_**B**=r2/(r1+r2)

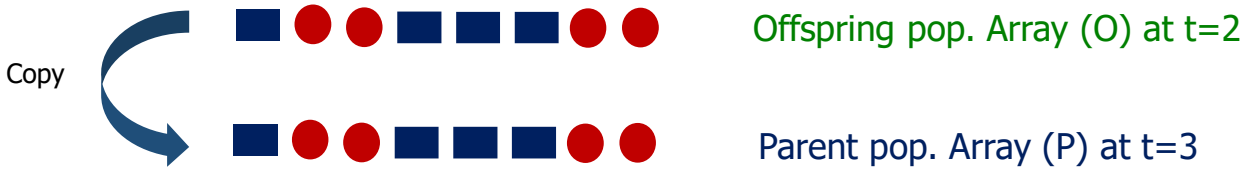
Pictorial representation of selection ***without*** mutation: Fixation of an ad/disadvantageous mutant

Wright- Fischer process: Evolution by updating whole population every generation



Repeat above steps to fill up the offspring pop. at t=2

Copy offspring array at t=2 to parent array such that it becomes the parent pop. For choosing offspring at t=3



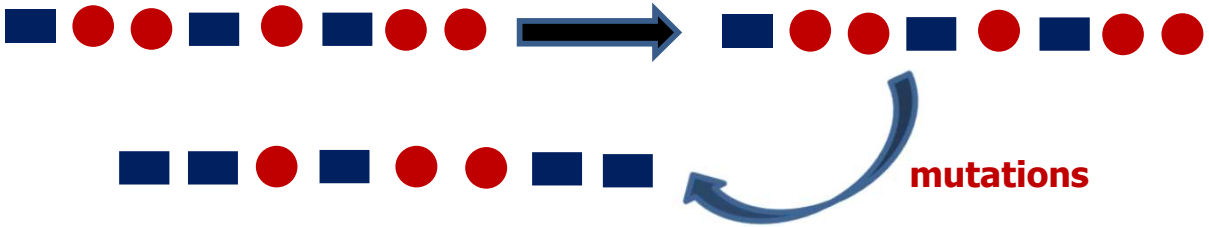
Pictorial representation of evolution via mutation and selection (Error Threshold Simulation)

Evolution by updating whole population every generation

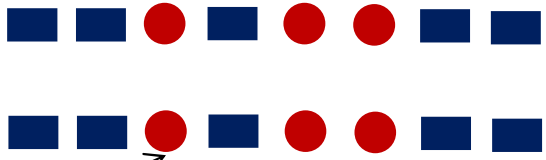
wright fischer

A-type B-type

Time=1 Copy parent pop. to offspring pop.



Time=1 Offspring pop. after **mutations** ➔



mutations

Time=1 Pick a Random member

Suppose RN=3

Offspring pop. Array (O)

B picked for reproduction

Not OK

Compare with fitness for reproduction: Pick RN between 0&1. If RN< normalized_fitness_of_B, then OK

normalized_fitness_of_B = f1/(f0+f1)

OK

B becomes the second member of the Offspring pop. at t=2

Time=2

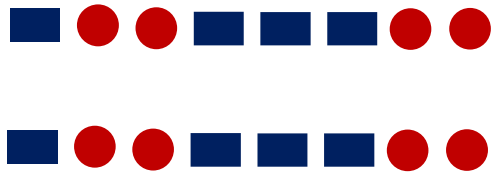
Parent pop. Array (P)

B becomes the second member of the Offspring pop. at t=2

Repeat above steps to fill up the parent pop. at t=2

Copy offspring array at t=2 to parent array such that it becomes the parent pop. For choosing offspring at t=3

Copy



Offspring pop. Array (O) at t=2

Parent pop. Array (P) at t=3